Naming an unknown molecule from its mass spectra

How the current public model for the Enveda CASMI 2026 Kaggle challenge works, and what our first strategy session taught us

October 18, 2026 · Competition runs Sep 14 – Dec 14, 2026 · $50k, top 5 paid

1 The problem

A tandem mass spectrometer weighs a charged molecule (the **precursor ion**), then breaks it apart by collisions with gas and weighs the pieces. Each **spectrum** is a list of peaks, (fragment m/z, intensity), where m/z is mass over charge. For each of ~400 hidden, natural-product-like test molecules we get 1–16 spectra (median 3), recorded on a Bruker timsTOF at 20/40/60 eV and with one of ten allowed adducts. We must return up to **25 ranked structures** (as SMILES).

- **Score (MRR@25):** 1/rank of the first correct guess (rank 1 → 1, rank 4 → 0.25, not in top 25 → 0), averaged over molecules. "Correct" = same atom connectivity (first block of the tautomer-canonical InChIKey; stereo ignored).
- **A noisy scoreboard:** the public leaderboard uses 31% of the test (~124 molecules). One molecule moving from miss to rank 1 shifts it by ~0.008; its standard error is ~0.016.
- **Data and rules:** 2.54M training spectra of ~275,810 structures from 11 public libraries; a Kaggle notebook, no internet, ≤ 9 h. For prizes, all data must be public and free (no NIST) and every pretrained weight reproducible from public code.
- **Standing (Oct 17):** top score 0.495; best public notebooks ~0.44–0.45; 3,486 teams.

Test class	What it means	Share*	How to solve
1 Known spectra	public reference spectra exist	16–28%	library search (MRR ~0.93)
2 Known structure	in PubChem or COCONUT, never measured publicly	20–45%	retrieve; rank among same-mass candidates
3 Novel	not in PubChem	39–52%	must be generated

*Community leaderboard-probe estimates; the true split is hidden.

Jargon in plain words

Adduct — the charged form that is measured, e.g. $[M+H]^+$ (molecule + proton) or $[M+Na]^+$. With it, the precursor m/z gives the neutral mass M.

Fragment — a piece left after bonds break; each peak is a fragment's m/z.

Isomers — different structures with the same formula, hence the same mass: the main source of confusion.

Fingerprint — a bit vector, one bit per substructure ("has an ester?"). *Tanimoto* = share of on-bits two fingerprints have in common.

Analog — a different molecule with a similar spectrum, often a small modification (e.g. +OH) of the unknown.

Forward model — a network that predicts a candidate's spectrum (structure → spectrum), used to check candidates.

ppm / ranker — mass error in parts per million / a model (here gradient-boosted trees: LightGBM, LambdaRank) that learns to order candidates from features.

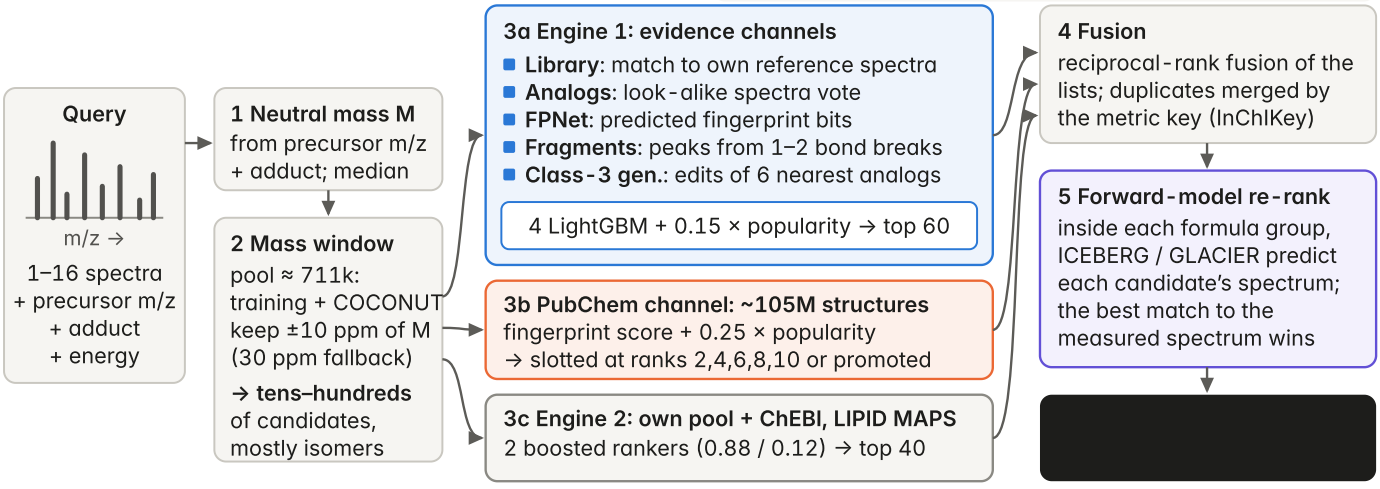


Figure 1. The current public model (community "E006" lineage; the notebook we forked scores 0.450). Mass turns ~711k structures into a short list of mostly isomers; three engines score it with complementary evidence; fusion and forward models produce the final top 25.

2 How the current public model works

The best public notebook (we forked dmitriigluzdov's "From Spectra to Structures") stacks many community contributions into one **retrieve-then-rank** pipeline (Fig. 1):

3. **PubChem channel:** fingerprint models screen ~105M PubChem structures in the window. Proposals are promoted to rank 1 when the fingerprint margin is large (> 150) or the structure is well documented, and skipped when a strong library match exists (similarity ≥ 0.9).

1. **Weigh and shortlist.** M is computed from each spectrum's precursor m/z and adduct (median over spectra). Every pool structure within 10 ppm is a candidate. Mass alone cannot separate isomers, so the spectra must decide.
2. **Engine 1 collects evidence per candidate:** *library* = spectral-entropy similarity to the candidate's own reference spectra (solves class 1); *analog propagation* = training spectra that resemble the query after a mass shift vote for candidates structurally similar to them (Tanimoto); *FPNet*, a transformer over peaks + metadata, predicts the fingerprint and candidates with agreeing bits score high; *fragmentation* = share of peak intensity explained by breaking 1–2 bonds (MetFrag-lite) plus a learned fragment scorer. Four LightGBM boosters combine ~100 features; popularity (log PubChem substance + literature counts) is then added.
4. **Engine 2** builds an independent list; **fusion** rewards candidates ranked high by either list; **forward models** break ties among isomers by predicting each one's spectrum.

History: ~0.34 (mid-Sep, 4-channel retrieval + ranker) → 0.40 (engine fusion) → 0.44 (E006 fusion + forward-model re-rank) → 0.45 (tuned PubChem promotion, deeper engine-2 trees).

Weak spots:

- almost no class-3 hits, though class 3 may be the largest class;
- many errors are the wrong same-formula isomer;
- most refinements were accepted on the noisy 124-molecule board without cross-validation;
- several bundled weights have uncertain licence or reproducibility (some CC-BY-NC-SA; one ranker-row file of unpublished construction): a prize risk.

3 Our strategy: what we tried, what we learned

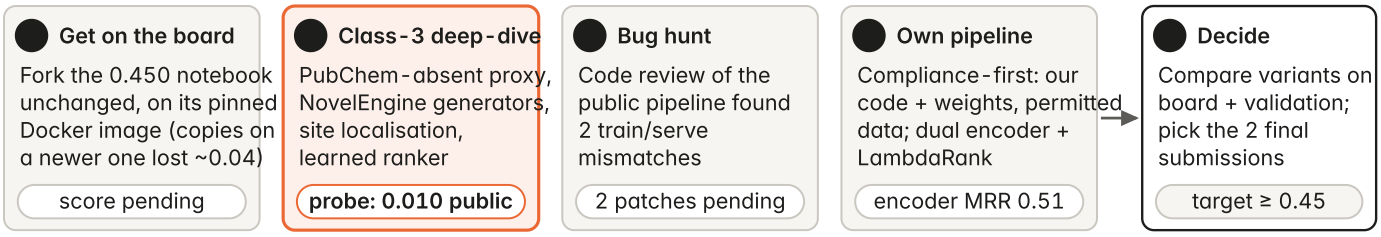


Figure 2. Our five tracks (session of Oct 17–18). A–D ran in parallel and feed the decision in E. Track B (orange) produced the main lesson.

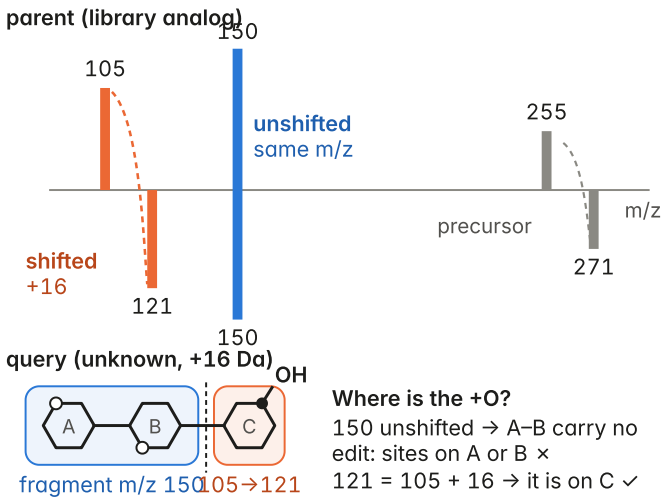
A — Get on the board. We submitted the 0.450 notebook unchanged and kept its pinned Docker image: teams that re-ran copies on a newer image lost ~0.04. (Score pending.)

B — Class 3: chasing the largest untapped share

We cannot see the hidden novel molecules, so we built a stand-in. Matching training structures against all 106M PubChem connectivity keys left **16,917 real, measured molecules that are not in PubChem**: our class-3 proxy. Insight: 45% of them split at one amide or ester bond into two PubChem-known pieces (N-acyl amino acids, bile-acid conjugates, fatty-acid esters), i.e. "hypothesised" natural products built from known blocks. Our **NovelEngine** has two generators that only emit structures absent from PubChem, COCONUT and training:

- **Analog derivatives:** find training spectra similar to the query by *modified cosine* (which also matches peaks shifted by the mass difference Δ), then apply edits (hydroxylation, methylation, glycosylation, chain extension, …) whose mass change equals Δ. From the top-20 analogs this reaches the answer for 24% of the proxy.
- **Conjugates:** join a known acid with a known amine or alcohol (amide/ester/glycoside) whose masses add up; rank by diagnostic fragment ions of each half.

Site localisation (our scoring idea, Fig. 3) decides where an edit sits. Scoring each candidate site by peak consistency raised derivative MRR on the proxy from 0.061 to 0.107; a learned LambdaRank ranker raised proxy MRR from 0.124 to 0.269.



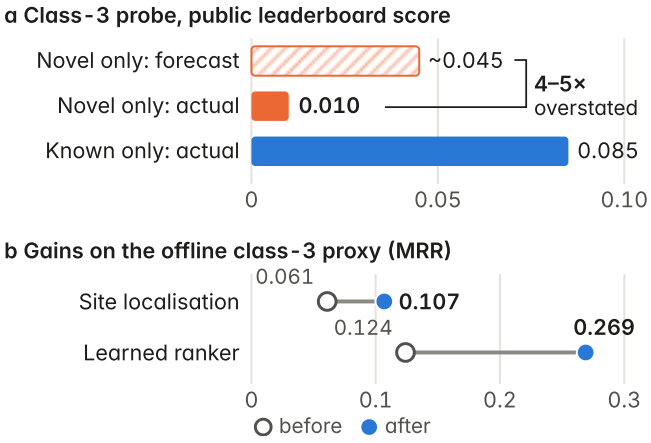


Figure 4. Proxy versus reality. (a) Submitting only novel candidates scored 0.010, against ~0.045 forecast from the proxy; the same engine's known-only list scored 0.085. (b) The proxy gains behind the forecast. Note the scales differ: (a) is the public test, (b) the proxy only.

The lesson we put centre stage

The proxy overstated class-3 reach ~4–5×, so class 3 is worth at most ~0.01 as an add-on, about one molecule moved to rank 1. The known-only probe (0.085) shows the engine works; the novel part barely pays. **A proxy is a hypothesis about the test: validate it against the leaderboard early, before investing in it.** And since the board itself is noisy (±0.016), we also need offline validation that mimics the test (track D).

C — Bug hunt. A subagent code review found two **train/serve mismatches** (a feature computed differently in training and at prediction time): (1) the "number of peaks" ranker feature uses raw test peaks (median 230) but was trained on cleaned peaks (median 52); (2) hidden-test positive-mode m/z values carry an electron-mass correction (−0.000548 Da) that the enveda-180 training spectra lack. This shifts mass-error features by ~1.6 ppm at the median mass, large next to the ~2.5 ppm typical gap between competing formulas. Two patched variants are submitted (scores pending).

D — Compliance-first own pipeline. For prize eligibility every model must be ours and reproducible. Pool: 726,198 structures (training + COCONUT Oct-2026). Channels: library entropy similarity, analog propagation, fragment explanation, COCONUT documentation priors (organisms, literature, synonyms) and our own **dual encoder**: peaks binned at 1 Da +

Figure 3. Site localisation (toy numbers). Top: a library analog ("parent", up) mirrored against the query (down), which is 16 Da heavier (+O). The parent peak at 150 reappears unshifted; the one at 105 reappears at 121. Bottom: so the fragment A–B is unmodified and the +O sits on ring C (filled dot) — candidate sites on A or B (open dots) are inconsistent.

neutral losses + adduct/energy/instrument → MLP → embedding, and 2,215 fingerprint bits → MLP → embedding, trained to put the true structure closer than up to 64 same-mass "hard negatives" (contrastive loss) while also predicting fingerprint bits. After one CPU epoch it ranks held-out structures among same-mass candidates with **MRR 0.51 (top-1 36%)**; longer runs are in progress. A LambdaRank ranker learns from simulated queries on 8,000 training structures (class 1: hide only the query spectra; class 2: hide all spectra of the structure), validated on 1,500 disjoint ones (250 enveda-np-examples, 750 natural products, 500 others), using the *measured* mass (the true one would leak the answer) plus the electron correction.

E — Next. Compare variants on the board; finish the own pipeline (target ≥ 0.45, fully compliant); add the class-3 engine only where the main list is weak; pick the two finals by validation + board. If hosts enforce the rules, strong compliant entries may outrank non-reproducible ones.